

WENXUAN GUO

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Education

University of Chicago, Booth School of Business

Sep. 2021 – Present

Ph.D. in Econometrics and Statistics

Chicago, IL

University of Chicago, Department of Statistics

Sep. 2019 – Jul. 2021

M.S. in Statistics

Chicago, IL

Fudan University, School of Mathematical Sciences

Sep. 2015 – Jul. 2019

B.S. in Mathematical Sciences

Shanghai, China

Research Interests

- Causal inference: optimal experimental design and randomization inference.
- Statistical inference and learning: finite-sample inference, online learning, and optimal transport.

Work in Progress

- **Wenxuan Guo**, Tengyuan Liang (2026). A Coupling Approach to Cost-Aware Online Portfolio Rebalancing.

Preprints

- **Wenxuan Guo**, Tengyuan Liang (2026). [Experimental Design when \$N\$ Equals One](#). *Submitted*.
- **Wenxuan Guo**, Panos Toulis, Yuhao Wang (2026). [Permutation Inference under Multi-way Clustering and Missing Data](#). *In preparation*.
- **Wenxuan Guo**, JungHo Lee, Panos Toulis (2025). [ML-assisted Randomization Tests for Complex Treatment Effects in A/B Experiments](#). *Under review at Neural Information Processing Systems (NeurIPS)*.
- **Wenxuan Guo**, Panos Toulis (2024). [Invariance-based Inference in High-Dimensional Regression with Finite-Sample Guarantees](#). *R&R at Journal of the American Statistical Association (JASA)*.

Publications

- **Wenxuan Guo**, Tengyuan Liang, Panos Toulis (2026). [Gaussianized Design Optimization for Covariate-Balancing in Randomized Experiments](#), *Journal of the Royal Statistical Society Series B (JRSS-B)*. *Proceedings of the ACM Conference on Economics and Computation (EC 2025)*, extended abstract.
- YoonHaeng Hur, **Wenxuan Guo**, Tengyuan Liang (2023). [Reversible Gromov-Monge Sampler for Simulation-Based Inference](#), *SIAM Journal on Mathematics of Data Science (SIMODS)*.
- **Wenxuan Guo**, YoonHaeng Hur, Tengyuan Liang, Christopher Ryan (2022). [Online Learning to Transport via the Minimal Selection Principle](#), *Conference on Learning Theory (COLT)*.

Academic Service

- Reviewers for major ML conferences (Neurips, ICML) and statistical journals.
- Teaching Assistantship
 - Time-series Analysis for Forecasting and Model Building (2025)
 - Applied Econometrics (Guest Lecturer, 2024 - 2026)
 - Business Statistics (2024 - 2026)
 - Modern Applied Optimization (2023)
 - Multivariate Data Analysis via Matrix Decompositions (2020)

Awards

- Silver Reviewer Award, ICML, 2026.
- Meritorious Winner, Mathematical Contest in Modeling, 2019.
- First Place, Contemporary Undergraduate Mathematical Contest in Modeling in China, 2017.

Talks

Permutation Inference under Multi-way Clustering and Missing Data

Econometrics & Statistics PhD Seminar, University of Chicago, Chicago, US.	03/2026
Chicago Chinese Social Research group (CCSRG), University of Chicago, Chicago, US.	02/2026

Gaussianized Design Optimization for Covariate-Balancing in Randomized Experiments

Joint Conference on Statistics and Data Science in China (JCSDS'2026), Guizhou, China.	07/2026
American Causal Inference Conference (ACIC'2026), Salt Lake City, US.	05/2026
Advances with Field Experiments Conference (AFE' 2025), University of Chicago, Chicago, US.	09/2025
Joint Statistical Meetings (JSM'2025), Nashville, US.	08/2025
ACM Conference on Economics and Computation (EC'2025), Stanford University, US.	07/2025
Econometrics & Statistics PhD Seminar, University of Chicago, Chicago, US.	04/2025

ML-assisted Randomization Tests for Complex Treatment Effects in A/B Experiments

American Causal Inference Conference (ACIC'2025), Detroit, US.	05/2025
Online Causal Inference Seminar (OCIS).	12/2024
– Discussant: Professor Xinran Li (University of Chicago)	

Invariance-based Inference in High-Dimensional Regression with Finite-Sample Guarantees

Joint Conference on Statistics and Data Science in China (JCSDS), Yunan, China.	07/2024
Econometrics & Statistics PhD Seminar, University of Chicago, Chicago, US.	01/2024